

C Programming: Program-First Learning Roadmap

A 180-program progression for a complete beginner — learn programming by solving programs, not by memorizing theory.

How to use this roadmap

For every program, you are required to understand every keyword, function, operator, variable, symbol and line before moving to the next program.

For every program, you must be able to explain:

- What every `#include` statement means.
- What every keyword means.
- What every variable and data type means.
- What every `printf()`, `scanf()`, or other function does.
- What every format specifier means.
- What every operator and symbol does.
- Why each pair of `{ }` is present.
- How the program executes line by line.
- What changes when the input changes.
- What happens if a line is removed or deliberately changed.

Level 1 — Absolute Basics

1. Print Hello World
2. Print your name, age and college name
3. Print a simple sentence using multiple `printf()` statements
4. Take a name as input and print it
5. Take an integer as input and print it
6. Take two integers and print their sum
7. Take two integers and perform addition, subtraction, multiplication and division
8. Calculate the square of a number
9. Calculate the cube of a number
10. Calculate the area of a circle
11. Calculate the area and perimeter of a rectangle
12. Convert Celsius to Fahrenheit
13. Swap two numbers using a third variable
14. Swap two numbers without using a third variable

Level 2 — Data Types, Format Specifiers & Expressions

15. Demonstrate `int`, `float`, `char` and `double`

16. Input and display different types of variables
17. Calculate simple interest
18. Calculate compound interest
19. Calculate average of three numbers
20. Calculate percentage of five subjects
21. Convert seconds into hours, minutes and seconds
22. Convert days into years, months and days
23. Convert rupees into paise
24. Calculate the total and average marks of a student

Level 3 — if, else and Conditions

25. Check whether a number is positive, negative or zero
26. Check whether a number is even or odd
27. Check whether a person is eligible to vote
28. Find the greater of two numbers
29. Find the greatest of three numbers
30. Check whether a year is a leap year
31. Check whether a character is a vowel or consonant
32. Check whether a character is uppercase or lowercase
33. Check whether a number is divisible by 5
34. Check whether a number is divisible by both 5 and 11
35. Check whether a student has passed or failed
36. Calculate grade based on marks
37. Check whether three sides can form a triangle
38. Determine whether a triangle is equilateral, isosceles or scalene
39. Find the roots of a quadratic equation without using library functions initially

Level 4 — switch

40. Make a simple calculator using switch
41. Display the day of the week using a number
42. Display the month name using a number
43. Create a menu-driven area calculator
44. Create a menu-driven unit converter
45. Create a simple menu-driven student result program

Level 5 — Loops

46. Print numbers from 1 to 10
47. Print numbers from 10 to 1
48. Print numbers from 1 to N
49. Print all even numbers from 1 to N

50. Print all odd numbers from 1 to N
51. Find the sum of numbers from 1 to N
52. Find the sum of even numbers from 1 to N
53. Find the sum of odd numbers from 1 to N
54. Print the multiplication table of a number
55. Print multiplication tables from 1 to 10
56. Find the factorial of a number
57. Count the digits of a number
58. Find the sum of digits of a number
59. Extract digits of a number from right to left
60. Extract digits of a number from left to right
61. Reverse a number
62. Check whether a number is a palindrome
63. Find the first and last digit of a number
64. Find the largest digit in a number
65. Find the smallest digit in a number
66. Count how many times a particular digit occurs
67. Check whether a number is an Armstrong number
68. Check whether a number is a perfect number
69. Check whether a number is a prime number
70. Print all prime numbers between two numbers
71. Find the HCF/GCD of two numbers
72. Find the LCM of two numbers
73. Generate the Fibonacci series
74. Check whether a number belongs to the Fibonacci series

Level 6 — Nested Loops & Patterns

75. Print a square of *
76. Print a rectangle of *
77. Print a right-angled triangle of *
78. Print an inverted right-angled triangle
79. Print a number triangle
80. Print an inverted number triangle
81. Print the pattern 1 22 333 4444
82. Print the pattern 1 12 123 1234
83. Print the pattern 1 11 121 1221
84. Print a pyramid of *
85. Print an inverted pyramid
86. Print a number pyramid

- 87. Print Floyd's triangle
- 88. Print Pascal's triangle
- 89. Print a diamond pattern
- 90. Print a hollow square
- 91. Print a hollow triangle
- 92. Print an X pattern using *

Level 7 — Functions

- 93. Create a function to add two numbers
- 94. Create a function to find the square of a number
- 95. Create a function to check even/odd
- 96. Create a function to find the greatest of three numbers
- 97. Create a function to calculate factorial
- 98. Create a function to check prime number
- 99. Create a function to reverse a number
- 100. Create a function to check palindrome
- 101. Create a function to calculate GCD
- 102. Create a function to calculate LCM
- 103. Create a menu-driven calculator using functions
- 104. Create a student-result program using separate functions

Level 8 — Arrays

- 105. Input and display elements of an array
- 106. Find the sum of array elements
- 107. Find the average of array elements
- 108. Find the largest element in an array
- 109. Find the smallest element in an array
- 110. Count even and odd elements in an array
- 111. Search for an element in an array
- 112. Count occurrences of an element
- 113. Reverse an array
- 114. Copy one array into another
- 115. Merge two arrays
- 116. Find the second-largest element
- 117. Sort an array in ascending order
- 118. Sort an array in descending order
- 119. Remove duplicate elements from an array
- 120. Insert an element into an array
- 121. Delete an element from an array

Level 9 — Strings

- 122. Input and display a string
- 123. Find the length of a string without using strlen()
- 124. Copy a string without using strcpy()
- 125. Compare two strings without using strcmp()
- 126. Concatenate two strings without using strcat()
- 127. Reverse a string
- 128. Check whether a string is a palindrome
- 129. Count vowels and consonants
- 130. Count digits, spaces and special characters
- 131. Convert lowercase characters to uppercase
- 132. Convert uppercase characters to lowercase
- 133. Count the frequency of each character
- 134. Remove spaces from a string
- 135. Find the largest word in a sentence
- 136. Count the number of words in a sentence

Level 10 — Pointers

- 137. Demonstrate the address-of (&) operator
- 138. Demonstrate the dereference (*) operator
- 139. Store and print the address of a variable
- 140. Change a variable's value using a pointer
- 141. Swap two numbers using pointers
- 142. Add two numbers using pointers
- 143. Find the largest number using pointers
- 144. Traverse an array using a pointer
- 145. Reverse an array using pointers
- 146. Manipulate a string using pointers

Level 11 — Structures

- 147. Store and display student information using a structure
- 148. Store information of 5 students
- 149. Find the student with the highest marks
- 150. Calculate student percentage using structures
- 151. Store employee information
- 152. Search for an employee by ID
- 153. Sort students according to marks
- 154. Create a structure containing another structure

Level 12 — Files

- 155. Create and write to a text file
- 156. Read and display a text file
- 157. Append data to an existing file
- 158. Count characters in a file
- 159. Count words in a file
- 160. Count lines in a file
- 161. Store student records in a file
- 162. Read student records from a file
- 163. Search for a student in a file
- 164. Update a student's record in a file

Level 13 — Small Projects

- 165. Simple calculator
- 166. Number guessing game
- 167. Rock-paper-scissors game
- 168. Simple quiz program
- 169. Student marks management system
- 170. Student record management system
- 171. Employee record management system
- 172. Simple banking system
- 173. Library management system
- 174. Contact management system
- 175. ATM simulation
- 176. Tic-Tac-Toe
- 177. Mini billing system
- 178. Inventory management system
- 179. Simple login system
- 180. Menu-driven mini database using files